DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

TITLE: Intrusion Detection System (IDS)

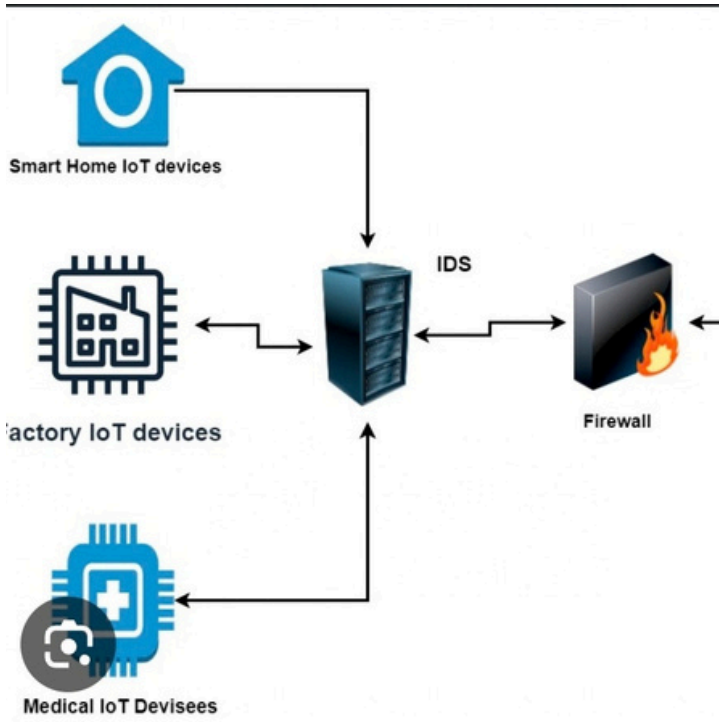


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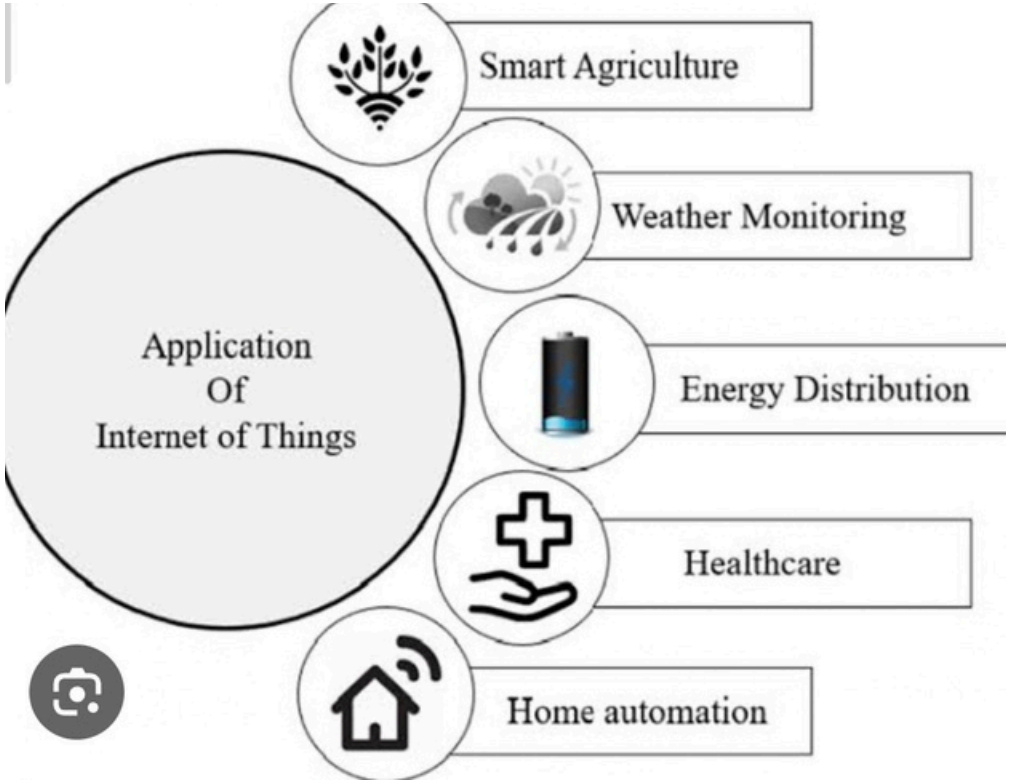
INTRODUCTION

- An Intrusion Detection System (IDS) is a security system designed to detect unauthorized entry or movement.
  This project uses a laser beam and LDR sensor as the intrusion-detection mechanism.
  Under normal conditions, the laser continuously falls on the LDR.
  When a person or object crosses the protected path, the beam is interrupted and the sensor output changes.
  The control circuit detects the change and activates a buzzer to provide an immediate alert.



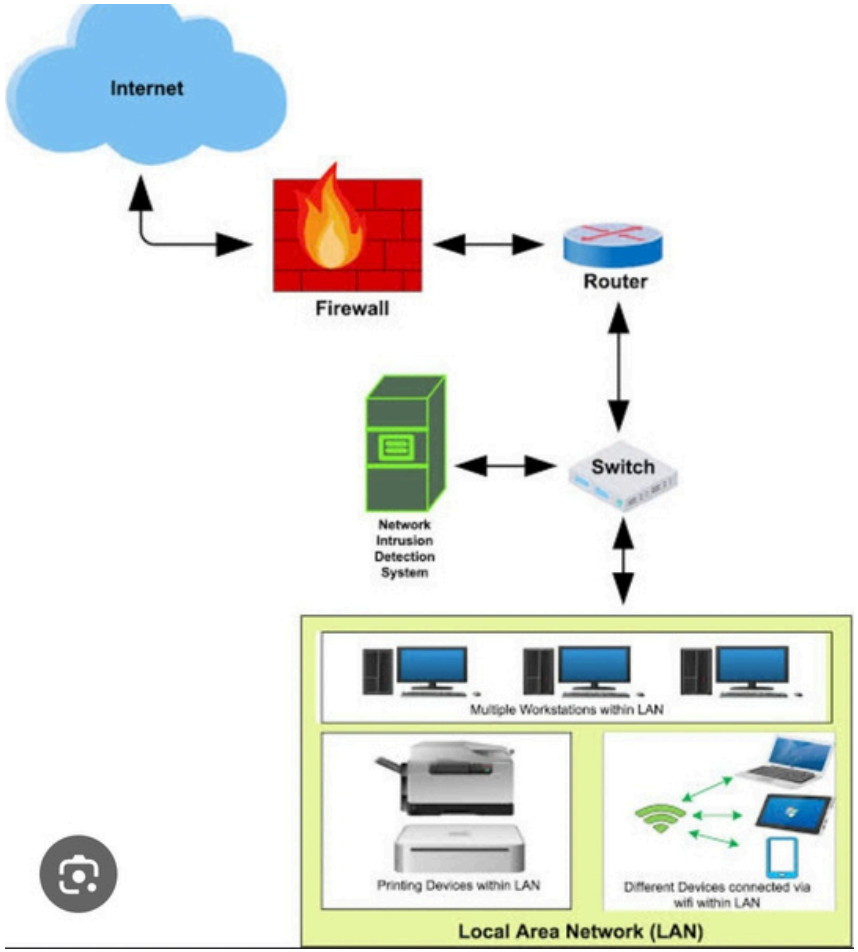
ABSTRACT

- A simple, low-cost Intrusion Detection System designed to detect unauthorized movement across a protected area.
  A laser beam is directed toward an LDR, which continuously receives light under normal conditions.
  When an intruder crosses the detection path, the laser beam is interrupted and the LDR output changes.
  A control circuit or microcontroller detects this change and activates a buzzer as an immediate intrusion alert.
  The system is economical, easy to construct and suitable for doors, entrances, laboratories and restricted areas.



LITERATURE SURVEY

- Traditional intrusion detection systems use physical switches, cameras, PIR sensors and other motion sensors.
  Infrared and light-based sensing techniques are widely used for detecting movement across protected zones.
  LDR-based detection circuits provide a low-cost method for sensing changes in light intensity.
  Laser beams provide a narrow and well-defined detection path for a focused security zone.
  Microcontrollers such as Arduino can process sensor signals and control alarms.
  Advanced IDS solutions can be improved using multiple sensors, wireless alerts, cameras and mobile notifications.



PROBLEM STATEMENT

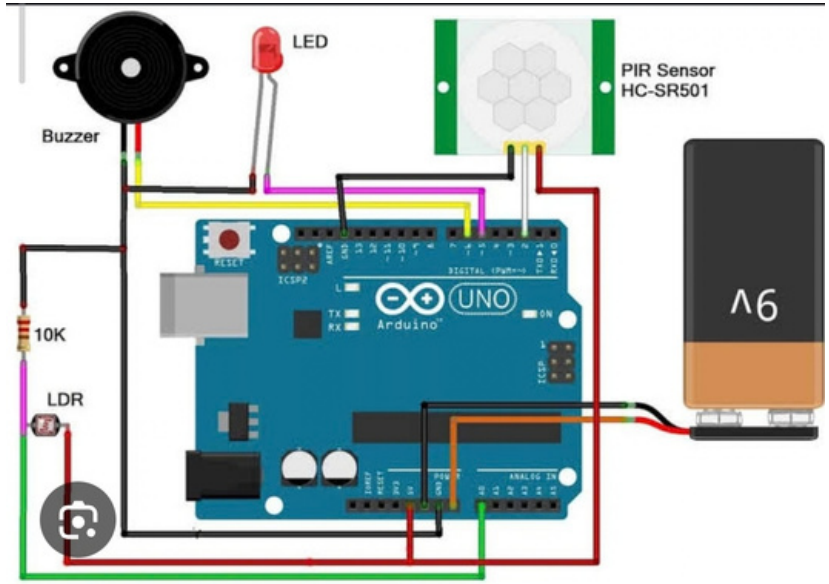
- Conventional security systems can be costly and may require continuous monitoring.
  There is a need for a simple, low-cost and easily implementable Intrusion Detection System.
  The proposed IDS monitors a predefined protected path using a laser beam and LDR sensor.
  When the detection path is interrupted, the system immediately activates an alarm to alert the user.

OBJECTIVES

The main objectives of the Intrusion Detection System (IDS) are:
To design a low-cost intrusion detection and alert system.
To detect unauthorized movement across a predefined protected path.
To use an LDR for detecting changes in light intensity.
To activate a buzzer when an intrusion is detected.
To provide a quick indication of a possible security breach.
To develop a simple system suitable for educational and small-scale security applications.

PROPOSED SYSTEM

- The proposed Intrusion Detection System consists of a laser module, LDR sensor, control circuit or microcontroller, buzzer and power supply.
  The laser continuously falls on the LDR during normal operation.
  When an intruder crosses the protected path, the laser beam is blocked and the LDR detects a reduction in light.
  The controller identifies the sensor change and activates the buzzer.
  The system therefore provides an immediate local alert when an intrusion is detected.



WORKING PRINCIPLE

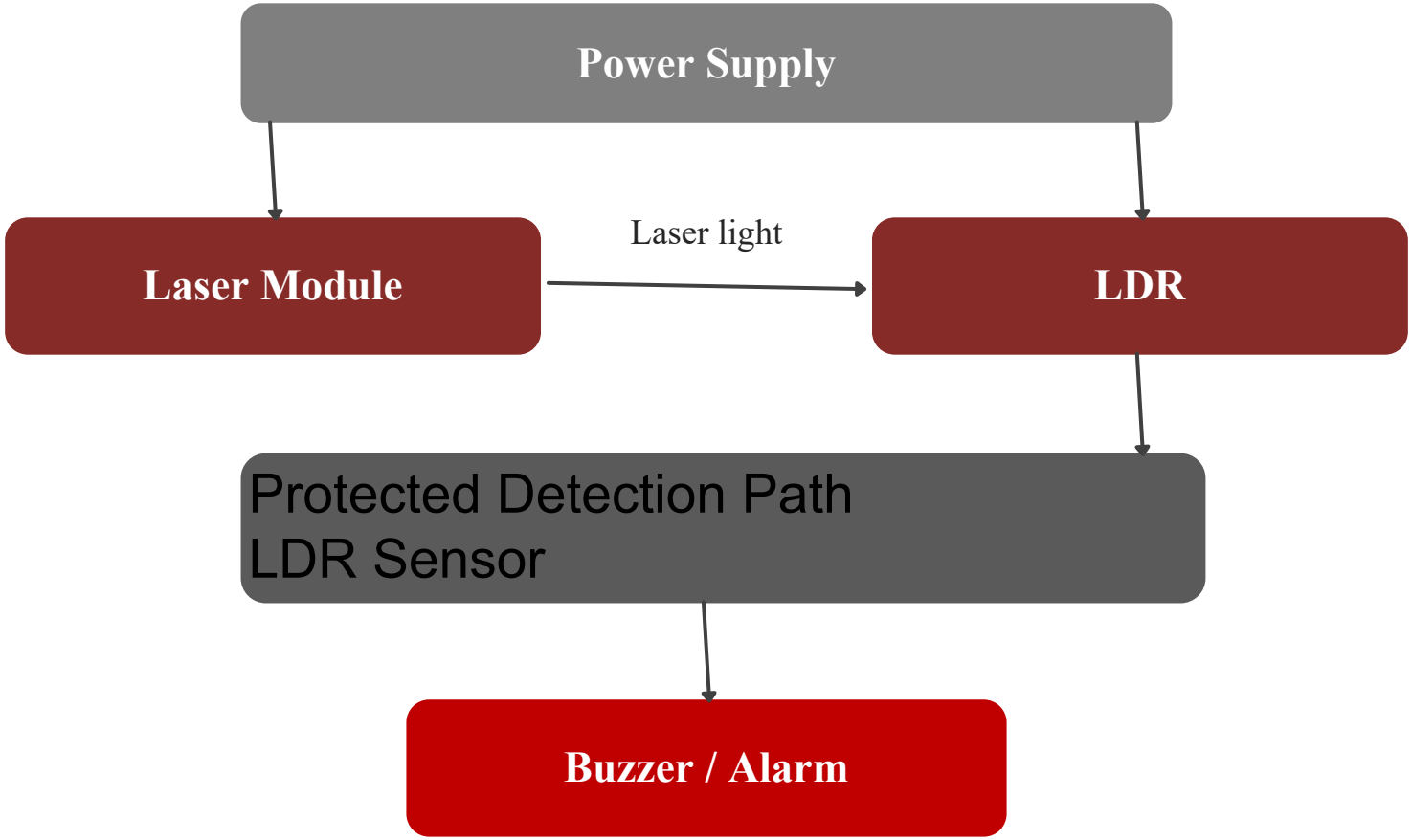
Normal operation



Intrusion detected



SYSTEM ARCHITECTURE



HARDWARE & SOFTWARE REQUIREMENTS

Hardware

- Hardware
  Laser diode / module
  LDR sensor
  Arduino Uno (if using a microcontroller)
  Buzzer
  Resistors
  Breadboard
  Jumper wires
  LED (optional)
  Battery / 5V power supply
  Transistor / relay (depending on circuit design)
  Software
  Arduino IDE
  Embedded C / C++ programming

Software

- Arduino IDE
- Embedded C / C++ programming

METHODOLOGY & IMPLEMENTATION

Step 1: Define the protected area – Select the path or entrance to be monitored.
Step 2: Laser setup – Fix the laser module so that its beam points toward the LDR.
Step 3: Sensor placement – Place the LDR directly in the laser path.
Step 4: Sensor interfacing – Connect the LDR to a voltage-divider/sensor circuit and control circuit or Arduino.
Step 5: Monitoring – Continuously monitor the LDR sensor value.
Step 6: Intrusion detection – Detect the change when the laser beam is blocked.
Step 7: Alarm activation – Switch ON the buzzer when the intrusion condition is detected.
Step 8: Testing – Test normal, intrusion and restored conditions repeatedly.

RESULTS & DISCUSSION

- The prototype successfully detects interruption of the laser beam.

Condition	LDR Status	Alarm
Laser continuously falls on LDR	Light detected	**OFF**
Laser beam interrupted	Light reduced / not detected	**ON**
Laser restored	Light detected	**OFF**

- The prototype successfully detects interruption of the laser beam as an intrusion event.
  Condition | LDR Status | IDS Alarm
  Normal condition | Light detected | OFF
  Intrusion / beam interrupted | Light reduced / not detected | ON
  Protected path restored | Light detected | OFF
  The system provides a quick local indication of intrusion. Performance depends on correct laser-LDR alignment and suitable ambient-light conditions.

PROGRAM WORKING

1. Read the LDR sensor value continuously using analogRead().
   Compare the sensor value with a preset detection threshold.
   When the protected beam is interrupted, the LDR reading changes.
   If the reading crosses the threshold, the Arduino identifies a possible intrusion.
   The controller switches the buzzer ON.
   When the laser beam is restored, the buzzer is switched OFF.
   The sensor value can also be displayed through the Serial Monitor for testing.

OUTPUT / TEST RESULTS

Condition	Event	LDR Status	Buzzer
Normal condition	Laser falls on LDR	Light detected	**OFF**
Beam interrupted	Object blocks laser	Light reduced / not detected	**ON**
Laser restored	Laser reaches LDR again	Light detected	**OFF**

- Condition | Event | LDR Status | Buzzer
  Normal condition | Protected path clear | Light detected | OFF
  Intrusion detected | Object blocks laser | Light reduced / not detected | ON
  System restored | Laser reaches LDR again | Light detected | OFF
  Expected result: the sensor value changes when the protected beam is blocked, allowing the controller to identify the intrusion and activate the alarm.

FUTURE SCOPE

The Intrusion Detection System can be further improved by:
Adding GSM / Wi-Fi notifications to send intrusion alerts to a mobile phone.
Using multiple laser beams or multiple sensors for larger protected areas.
Adding an LCD / OLED display for system status.
Adding a camera for image / video capture after an intrusion event.
Developing an IoT-based remote monitoring system.
Adding battery backup for continuous operation.
Using a mobile application for remote monitoring and alert history.

CONCLUSION

- The Intrusion Detection System (IDS) demonstrates a simple, low-cost security solution using a laser, LDR sensor and buzzer.
  The controller continuously monitors the sensor, compares the reading with a threshold and controls the alarm.
  The prototype demonstrates three basic states: normal protected condition, intrusion detected and system restored.
  The project can be extended with wireless notifications, multiple detection zones, cameras, displays and IoT-based monitoring.



THANK YOU

Intrusion Detection System (IDS)